# Adam Boesky

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## Education

**Astrophysics PhD Student, Harvard University** *Cambridge, MA* *Aug 2026 – Present*

**BA, Harvard University** *Cambridge, MA* *Aug 2021 – May 2025*

***Majors:*** Astrophysics, Mathematics ***Honors:*** *cum laude*

***Relevant Coursework:*** General Relativity, Information Theory, Machine Learning for Astrophysicists, Advanced Scientific Computing, Differential Geometry, Dynamical Systems

## Professional & Research Experience

**Center for Astrophysics | Harvard & Smithsonian** *Cambridge, MA*

*Berger Cosmic Transient Group* *Jun 2022 – Present*

- Investigated the evolution and demographics of black holes and neutron stars with population synthesis simulations of massive binary stars
- For senior thesis: created catalog of long-duration transients with novel source extraction methods; studying the detectability of long-duration transients with LSST and *Roman*

*Villar Time-Domain Astronomy Data Lab* *Aug 2023 – Present*

- Developed SPLASH – a rapid photometric machine learning pipeline that classifies supernovae using their inferred host galaxy's properties; preliminary website here

**ProDex Labs** *New York, NY* *Mar 2025 – Mar 2026*

*Founding Engineer*

- Developed a general purpose discrete-event simulation engine for manufacturers to create digial twins of production processes
- Designed and implemented optimizers for advanced planning and scheduling module
- Integrated AI agent into application modules, enhancing user analysis capabilities and accelerating the user experience by $> 100\times$
- Led the development of several application modules; managed teams of engineers

**SpaceX** *Los Angeles, CA* *Summers 2023, 2024*

*Guidance, Navigation, & Control Engineering Intern*

- Developed physical and empirical models for high-fidelity Monte Carlo simulation of the Falcon9 and Falcon Heavy rockets
- Drove operational changes to promote launch vehicle reliability and performance
- Automated day of launch operations and implemented tools to accelerate launch cadence
  - Conducted study to assess the value of mission-specific analyses required for launch

**Harvard School of Engineering & Applied Sciences** *Cambridge, MA*

*Harvard C3 Policing Research Team* *Jun 2020 – Aug 2022*

- Quantified impact of novel Counter Criminal Continuum (C3) policing method; presented results to Massachusetts government and law enforcement officials
- Identified statistically significant positive trends in business, crime, education, health, and

housing data

## Open-Source Software

**astro_SPLASH** — *Lead Developer* (1 stars / 1 forks)
Supernova classification Pipeline Leveraging Attributes of Supernova Hosts (SPLASH): A machine learning pipeline that classifies supernovae by first inferring their host's properties
**ztforce** — *Lead Developer* (1 stars / 0 forks)
PSF photometry on ZTF science images, calibrated against PanSTARRS
**COMPAS** — *Developer* (91 stars / 84 forks)
COMPAS rapid binary population synthesis code [docs]
**Prost** — *Developer* (9 stars / 5 forks)
A probabilistic host-galaxy association code.
**HOOTSim** — *Lead Developer* (2 stars / 0 forks)
N-Body Simulator

## Teaching & Mentorship

*Teaching Assistant* — Harvard Physics 143a: Quantum Mechanics I Spring 2024
*Teaching Assistant* — Harvard Physics 12b: Electromagnetism and Statistical Physics Fall 2023

I am currently mentoring and have mentored several students on a variety of projects including: **High schoolers:** Khushi Karthikeyan (Mar 2024 – Present; *2026 Regeneron Science Talent Search Finalist*), Meera Desawale (Jul 2024 – Present), Daniel Garrett-Metz (Apr 2021 – Jun 2021)

## Professional Service

*Referee* — The Astrophysical Journal 2026 – Present

## Honors, Awards, and Fellowships

NSF Graduate Research Fellowship Program, 2026
*Honorable Mention* NSF Graduate Research Fellowship Program, 2025
*Finalist* Hertz Fellowship, 2025
*Finalist* Rhodes Scholarship, 2025
*Finalist* Marshall Scholarship, 2025
Harvard Program for Research in Science and Engineering Fellow ($5000), 2022
Harvard College Research Program Fellow ($1000), 2022

## Talks & Presentations

### Oral

- Orange County Astronomers, Orange, CA Jan 2026

- American Physical Society April Meeting, Sacramento, CA Apr 2024
- Post-Pax-VIII, Cambridge, MA Aug 2022

**Poster**

- American Astronomical Society Meeting, National Harbor, MD Jan 2025
- Computational and Systems Neuroscience, Lisbon, Portugal Mar 2024
- Giant Magellan Telescope 2022 Science Meeting, Sedona, AZ Sep 2022

## Skills & Interests

**Programming:** (Expert) Python, C++, Java, Mathematica, JavaScript, SQL (Intermediate) R (Familiar) Julia

**Software:** Git/GitHub (>500 contributions, 20 repositories), Slurm, LaTeX, Linux, Bash, SAOImage DS9

**Miscellaneous:** Billiards, skiing, running, mixology, mountaineering, knitting, surfing, foosball